

ERDŐS PROBLEM 928: QUANTITATIVE COMPARISONS AND ORIGINAL FORMULATIONS

JI HO BAE

ABSTRACT. We deduce the density formula $\rho(1/\alpha)\rho(1/\beta)$ for the precise formulation of Erdős Problem 928 from OpenAI's joint Dickman law. Here ρ is the Dickman–de Bruijn function. For fixed $0 < \alpha, \beta < 1$ and real $X \geq 2$, we bound the symmetric difference between the counting sets in the problem and in the cited theorem, restricted to $2 \leq n \leq X$, by $\pi(X^\alpha) + \pi((X+1)^\beta)$, where π counts primes. This elementary estimate accounts for both the strict inequalities and the change in the second threshold from n^β to $(n+1)^\beta$. We also treat the strict formulation with both thresholds based at n and compare the original statements in Erdős's two 1976 papers and the 1978 Erdős–Pomerance paper. The analytic joint law is due to OpenAI; this note provides the finite comparison estimates and the precise correspondence among the source formulations.

1. INTRODUCTION

For an integer $n \geq 2$, let $P(n)$ be the largest prime factor of n , and set $P(1) = 1$. Write ρ for the Dickman–de Bruijn function: the continuous function on $[0, \infty)$ determined by

$$\rho(u) = 1 \quad (0 \leq u \leq 1), \quad u\rho'(u) = -\rho(u-1) \quad (u > 1).$$

The classical smooth-number limit is

$$(1) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{1 \leq n \leq X : P(n) \leq X^c\} = \rho(1/c) \quad (0 < c < 1);$$

see de Bruijn [2, p. 51, (1.1)–(1.2)]. We also use the standard properties $\rho(u) > 0$ for finite $u \geq 0$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$; see de Bruijn [2, p. 52, (1.8)] together with the defining equation. We use $\mathbb{N} = \{1, 2, \dots\}$. Counting variables are integers, whereas the cutoff X is real. Natural density is the limit of the ordinary, unweighted count up to X , divided by X .

Problem 928 in Bloom's collection [1] asks whether, for fixed $\alpha, \beta \in (0, 1)$, the integers satisfying

$$(2) \quad P(n) < n^\alpha, \quad P(n+1) < (n+1)^\beta$$

have a natural density. The accompanying discussion asks whether its value is $\rho(1/\alpha)\rho(1/\beta)$. The page cites Erdős's Manitoba and Debrecen papers [3, 4] and the paper of Erdős and Pomerance [5]. Section 3 compares their original statements.

Our analytic input is OpenAI's joint Dickman law [6, Theorem 1.1, p. 2]: for every fixed $\alpha, \beta \in (0, 1)$,

$$(3) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \leq n^\alpha, P(n+1) \leq n^\beta\} = \rho(1/\alpha)\rho(1/\beta).$$

OpenAI presents this theorem as a resolution of the Erdős–Pomerance joint Dickman conjecture. Section 11.3 of the same paper also establishes the versions with thresholds X^α, X^β and the corresponding upper-tail conditions. The present note connects these analytic results to the precise conventions in the cited sources. Proposition 1 quantifies the changes needed for (2) and for Erdős's strict formulation; Section 3 explains how the other source statements follow.

2. FINITE COMPARISON ESTIMATES

Fix $\alpha, \beta \in (0, 1)$. For $X \geq 2$, define

$$\begin{aligned}\mathcal{A}_{\alpha,\beta}(X) &= \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < (n+1)^\beta\}, \\ \mathcal{B}_{\alpha,\beta}(X) &= \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) \leq n^\alpha, P(n+1) \leq n^\beta\}, \\ \mathcal{C}_{\alpha,\beta}(X) &= \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < n^\beta\}.\end{aligned}$$

The sets $\mathcal{A}$ and $\mathcal{C}$ correspond, respectively, to Problem 928 and condition (1) in Erdős's Manitoba paper [3, p. 25]; $\mathcal{B}$ is the set counted in (3). Let $\pi(y) = \#\{p \leq y : p \text{ is prime}\}$, and write $S \triangle T$ for the symmetric difference of two sets.

Proposition 1 (Comparison of conventions). *For every $\alpha, \beta \in (0, 1)$ and every real $X \geq 2$,*

$$(4) \quad \#(\mathcal{A}_{\alpha,\beta}(X) \triangle \mathcal{B}_{\alpha,\beta}(X)) \leq \pi(X^\alpha) + \pi((X+1)^\beta),$$

$$(5) \quad 0 \leq \#\mathcal{B}_{\alpha,\beta}(X) - \#\mathcal{C}_{\alpha,\beta}(X) \leq \pi(X^\alpha) + \pi(X^\beta).$$

Proof. In this proof we suppress the subscripts and the argument X . Suppose first that $n \in \mathcal{B} \setminus \mathcal{A}$. Since $P(n+1) \leq n^\beta < (n+1)^\beta$, the second condition defining $\mathcal{A}$ holds. Thus the first condition fails, giving $P(n) = n^\alpha$. The prime $p = P(n) \leq X^\alpha$ determines the only possible value $n = p^{1/\alpha}$. Hence $\#(\mathcal{B} \setminus \mathcal{A}) \leq \pi(X^\alpha)$.

Now suppose that $n \in \mathcal{A} \setminus \mathcal{B}$. The first condition defining $\mathcal{B}$ holds, and failure of the second gives

$$n^\beta < P(n+1) < (n+1)^\beta.$$

Because $\beta > 0$, the intervals $(m^\beta, (m+1)^\beta)$ for integers $m \geq 2$ are pairwise disjoint. The map $n \mapsto P(n+1)$ is therefore injective on $\mathcal{A} \setminus \mathcal{B}$, with values among the primes less than $(X+1)^\beta$. Thus $\#(\mathcal{A} \setminus \mathcal{B}) \leq \pi((X+1)^\beta)$. Adding the two bounds proves (4).

For the remaining comparison, $\mathcal{C} \subseteq \mathcal{B}$, and every $n \in \mathcal{B} \setminus \mathcal{C}$ satisfies $P(n) = n^\alpha$ or $P(n+1) = n^\beta$. The first equality contributes at most $\pi(X^\alpha)$ integers, as above. For the second equality, $n = P(n+1)^{1/\beta}$ is determined by a prime at most X^β , giving at most $\pi(X^\beta)$ integers. The union bound proves (5). $\square$

Corollary 2 (Problem 928 and Erdős's strict formulation). *For every fixed $\alpha, \beta \in (0, 1)$,*

$$(6) \quad \lim_{X \rightarrow \infty} \frac{\#\mathcal{A}_{\alpha,\beta}(X)}{X} = \lim_{X \rightarrow \infty} \frac{\#\mathcal{C}_{\alpha,\beta}(X)}{X} = \rho(1/\alpha)\rho(1/\beta).$$

Thus the density in Problem 928 and the density $d(\alpha, \beta)$ asked for in Erdős's Manitoba paper [3, p. 25] both exist and equal the predicted product.

Proof. For finite sets, $|\#S - \#T| \leq \#(S \triangle T)$. Since $\pi(y) \leq [y]$, both upper bounds in Proposition 1 are $O(X^\alpha + X^\beta) = o(X)$. Dividing the comparison bounds by X and using (3) gives both limits in (6). $\square$

The comparison estimates are independent of the joint Dickman law: they bound the discrepancy caused by the changes of convention for fixed α, β . The asymptotic evaluation of the count comes from (3); the comparison bounds do not give a rate of convergence in that law.

3. COMPARISON WITH THE ORIGINAL SOURCES

We call a threshold *moving* if it depends on the integer being counted, and *fixed* if it remains constant within the counting range. Thus n^α and $(n+1)^\beta$ are moving thresholds, whereas X^α and X^β are fixed thresholds. The latter still vary with the counting endpoint. Put

$$a(0) = 1, \quad a(t) = 1 - \rho(1/t) \quad (0 < t \leq 1).$$

Table 1 places the reference theorem beside the four source formulations. The final column records the normalized limits and the results from which they follow. The subsections below explain the changes in inequalities, thresholds, and counting ranges.

TABLE 1. Reference theorem and source formulations. In each row, N counts the integers satisfying the listed conditions. Exponents are fixed: $0 < \alpha, \beta < 1$, $\gamma > 1$, and $0 \leq t, s \leq 1$. Here $c_\gamma = \rho(\gamma)$. Unless indicated otherwise, $2 \leq n \leq X$ and $X \rightarrow \infty$.

Formulation and source	Conditions	Limit and justification
Reference theorem OpenAI [6, Thm. 1.1, p. 2]	$P(n) \leq n^\alpha$ $P(n+1) \leq n^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ Joint Dickman law (3)
Problem 928 Bloom [1]; Erdős's independence statement [3, p. 25]	$P(n) < n^\alpha$ $P(n+1) < (n+1)^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ This note: Proposition 1 and Corollary 2 (set $\mathcal{A}$)
Erdős's strict formulation Manitoba paper, condition (1) [3, p. 25]	$P(n) < n^\alpha$ $P(n+1) < n^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ This note: Proposition 1 and Corollary 2 (set $\mathcal{C}$)
Fixed-threshold pair Erdős (Debrecen), $r = 2$ [4, pp. 272–273]	$P(n+1) \leq k$ $P(n+2) \leq k$ $1 \leq n < k^\gamma, k \rightarrow \infty$	$\frac{N}{k^\gamma} \rightarrow \rho(\gamma)^2 = c_\gamma^2$ OpenAI [6, (11.7), p. 80], with the change of variables in Section 3.2
Fixed upper tails Erdős–Pomerance [5, p. 311]	$P(n) \geq X^t$ $P(n+1) \geq X^s$	$\frac{N}{X} \rightarrow a(t)a(s)$ OpenAI [6, (11.8) and following paragraph, p. 80], including equality and endpoints

3.1. Erdős's Manitoba paper. On page 25 of his Manitoba paper [3], Erdős expects independence for $P(n) < n^\alpha$ and $P(n+1) < (n+1)^\beta$. He then asks whether the integers satisfying condition (1), namely $P(n) < n^\alpha$ and $P(n+1) < n^\beta$, have a density $d(\alpha, \beta)$. The webpage therefore follows the inequalities in the independence statement, while condition (1) uses n^β as its second threshold. Corollary 2 establishes the density in both formulations.

Erdős also asks whether these strict inequalities have infinitely many solutions for each fixed pair $\alpha, \beta > 0$. The positive value in (6) gives infinitude. For parameters outside $(0, 1)$, the same conclusion follows by choosing smaller positive parameters inside that interval. This infinitude result was already available: as recorded on Bloom's webpage [1], it follows from Schinzel [7, Theorem 14, p. 231], applied to $(2n+1)^2 - 1 = 4n(n+1)$. The theorem gives a constant $K > 0$ and infinitely many integers $n \geq 3$ such that

$$P(n(n+1)) \leq n^{K/\log \log n}.$$

Since $K/\log \log n \rightarrow 0$, these integers satisfy Erdős's strict inequalities for every prescribed positive pair α, β , once n is sufficiently large.

3.2. Erdős's Debrecen paper. To distinguish the notation from that used above, write $\gamma > 1$ for the exponent denoted by α in Erdős's Debrecen paper [4, pp. 272–273]. Erdős recalls that the number of $m \leq k^\gamma$ satisfying $P(m) \leq k$ is $(c_\gamma + o(1))k^\gamma$. He then conjectures that, for each fixed integer $r \geq 1$, as k tends to infinity through positive integers,

$$(7) \quad \#\{1 \leq n < k^\gamma : P((n+1) \cdots (n+r)) \leq k\} = (c_\gamma^r + o(1))k^\gamma.$$

The marginal law (1) identifies $c_\gamma = \rho(\gamma)$.

For the two-variable case, we use OpenAI's fixed-threshold law [6, (11.7), p. 80]: for fixed $c, d \in (0, 1)$,

$$(8) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq m \leq X : P(m) \leq X^c, P(m+1) \leq X^d\} = \rho(1/c)\rho(1/d).$$

To obtain the case $r = 2$ of (7), put $X = k^\gamma$, $m = n+1$, and $c = d = 1/\gamma$. The identity $P(m(m+1)) = \max\{P(m), P(m+1)\}$ makes the prime-factor conditions exactly those in (8).

After this shift, the ranges $1 \leq n < X$ and $2 \leq m \leq X$ differ by at most one endpoint term. Its normalized contribution vanishes, so (8) gives (7) for $r = 2$.

Bloom's webpage [1] states a higher-order version with $P(n+i) \leq n^\alpha$ for $0 \leq i < r$. Its case $r = 2$ is precisely (3) with $\beta = \alpha$. The results used here concern two consecutive integers; they do not establish the assertion for $r \geq 3$.

3.3. Erdős and Pomerance. Erdős and Pomerance [5, p. 311] formulate independence using fixed thresholds and upper tails. For $0 \leq t, s \leq 1$, they ask whether

$$(9) \quad b(t, s) = \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \geq X^t, P(n+1) \geq X^s\}$$

exists, is continuous on $[0, 1]^2$, and equals $a(t)a(s)$. Their $a(t)$ is the corresponding one-variable upper-tail limit. The marginal law (1), with the equality cases and endpoints treated below, identifies their $a(t)$ with the function defined above.

For $0 < t, s < 1$, apply inclusion–exclusion to (8) and the two marginal laws. This gives the limit $a(t)a(s)$ for the strict upper-tail conditions $P(n) > X^t$, $P(n+1) > X^s$. Replacing n by $n+1$ in a marginal count changes its counting range by only $O(1)$ terms, so its limit is unchanged.

Allowing equality does not change the limit. For fixed $0 < t \leq 1$, equality $P(m) = X^t$ can occur only when X^t is a prime p . Then $p \mid m$, so at most $\lfloor (X+1)/p \rfloor = O(X^{1-t} + 1) = o(X)$ such $m \leq X+1$ occur.

At exponent 0 the upper-tail condition is automatic for $m \geq 2$. When just one exponent is zero, the remaining condition is evaluated by the marginal law. At exponent 1, the condition $P(m) \geq X$ in the range $2 \leq m \leq X+1$ forces $m \in [X, X+1]$, so at most two integers occur. Thus (9) holds at all endpoints as well. The continuity of ρ , together with $\rho(1) = 1$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$, shows that a is continuous on $[0, 1]$. Hence $b(t, s) = a(t)a(s)$ is continuous on the whole square.

OpenAI already gives this deduction, including the equality cases and endpoints, in [6, (11.8) and following paragraph, p. 80]. The purpose here is to identify its exact correspondence with the Erdős–Pomerance formulation [5]. Thus all four source formulations in Table 1 follow from the cited joint Dickman law and its fixed-threshold version, through the comparisons above.

REFERENCES

- [1] T. F. Bloom, *Erdős Problem #928*, accessed October 9, 2026.
<https://www.erdosproblems.com/928>.
- [2] N. G. de Bruijn, *On the number of positive integers $\leq x$ and free of prime factors $> y$* , Nederl. Akad. Wetensch. Proc. Ser. A **54** (1951), 50–60.
<https://pure.tue.nl/ws/files/1793853/597499.pdf>.
- [3] P. Erdős, *Problems and results on number theoretic properties of consecutive integers and related questions*, Proceedings of the Fifth Manitoba Conference on Numerical Mathematics (Winnipeg, 1975), Congressus Numerantium XVI, Utilitas Math., Winnipeg, 1976, pp. 25–44.
https://users.renyi.hu/~p_erdos/1976-39.pdf.
- [4] P. Erdős, *Problems and results on consecutive integers*, Publ. Math. Debrecen **23** (1976), 271–282.
<https://doi.org/10.5486/PMD.1976.23.3-4.15>.
- [5] P. Erdős and C. Pomerance, *On the largest prime factors of n and $n+1$* , Aequationes Math. **17** (1978), 311–321.
https://users.renyi.hu/~p_erdos/1978-29.pdf.
- [6] OpenAI, *The joint Dickman law for consecutive integers*, OpenAI Math Release preprint, September 24, 2026, commit `fd4aeeb2ee4f`.
<https://github.com/openai/math/blob/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/preprints/The-joint-Dickman-law-for-consecutive-integers-September-24-2026/paper.pdf>.
- [7] A. Schinzel, *On two theorems of Gelfond and some of their applications*, Acta Arith. **13** (1967), 177–236.
<https://doi.org/10.4064/aa-13-2-177-236>.